



Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

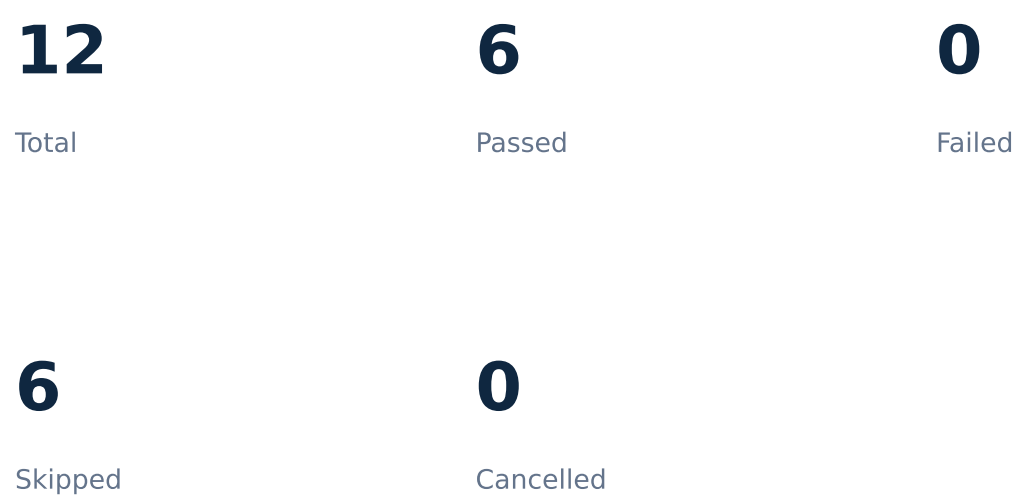
20260813_213606_4ddcf2

Date

2026-08-13T20:36:06.638340+00:00

Executive Summary

20260813_213606_4ddcf2



Overall measured availability: 71.69%

During this campaign, 6 of 12 scenarios passed, 0 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	1.813	13.719	True	True	9.488	35.833	11	21.43	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate failover routes require authentication, database persistence required	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	24.25	82.901	N/A	100.0	PASS	Authentication, database persistence required	N/A
memory	swarm	N/A	N/A	N/A	Not Required	97.519	747.738	N/A	100.0	PASS	Authentication, database persistence required	N/A
network-partition	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Authentication, database persistence required	N/A
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Authentication, database persistence required	N/A
container-kill	kubernetes	24.265	36.297	True	True	1671.265	2466.778	13	13.33	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Authentication, database persistence required	N/A
cpu	kubernetes	79.422	15.312	True	True	343.877	1198.742	4	96.23	PASS	Authentication, database persistence required	N/A
memory	kubernetes	46.516	3.172	True	True	143.774	579.239	1	99.14	PASS	Authentication, database persistence required	N/A
network-partition	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Authentication, database persistence required	N/A
latency	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Authentication, database persistence required	N/A

swarm — container-kill

State: **Pass**
Detection Time (s): 1.813

Recovery Time (s): 13.719

HTTP Success: N/A

HTTP Failures: 11

Availability (%): 21.43

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

Status: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

swarm — latency

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — container-kill

State: PASS
Detection Time (s): 24.265

Recovery Time (s): 36.297

HTTP Success: N/A

HTTP Failures: 13

Availability (%): 13.33

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

State: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: PASS
Detection Time (s): 79.422

Recovery Time (s): 15.312

HTTP Success: N/A

HTTP Failures: 4

Availability (%): 96.23

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

Status: Pass
Detection Time (s): 46.516

Recovery Time (s): 3.172

HTTP Success: N/A

HTTP Failures: 1

Availability (%): 99.14

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — network-partition

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — latency

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

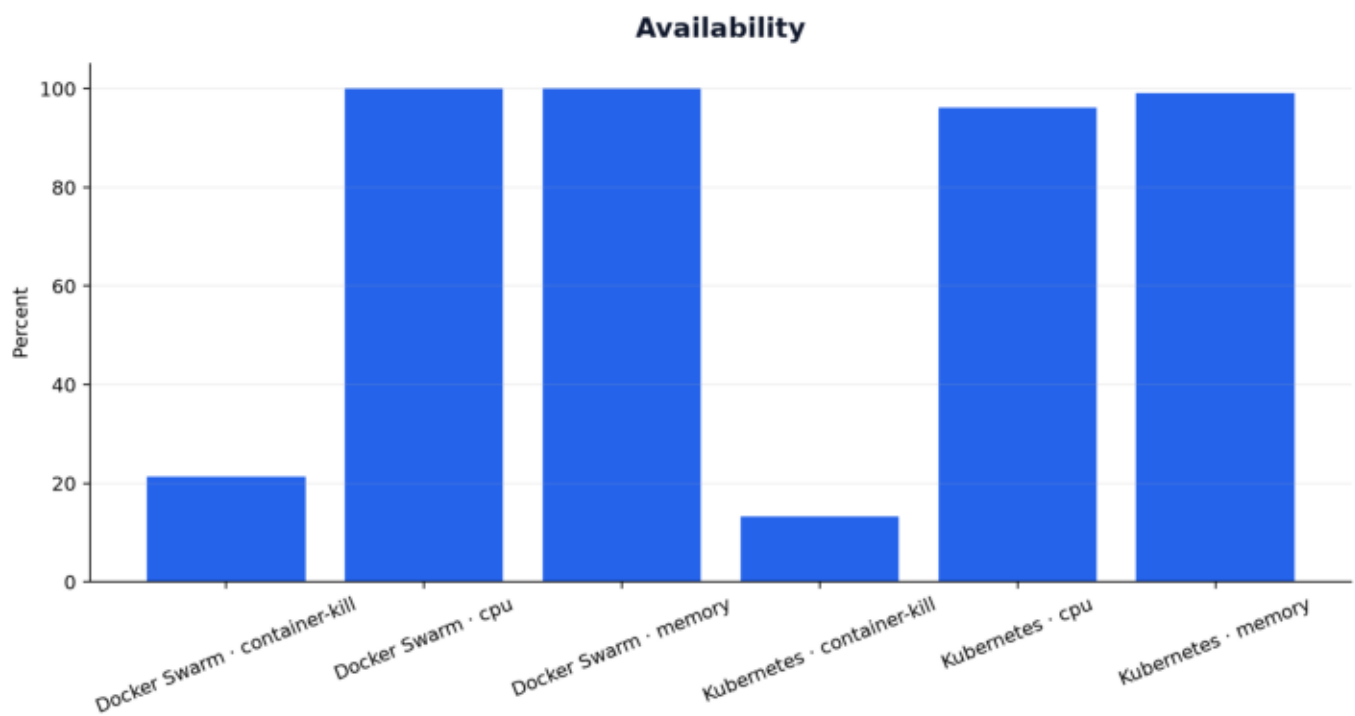
Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

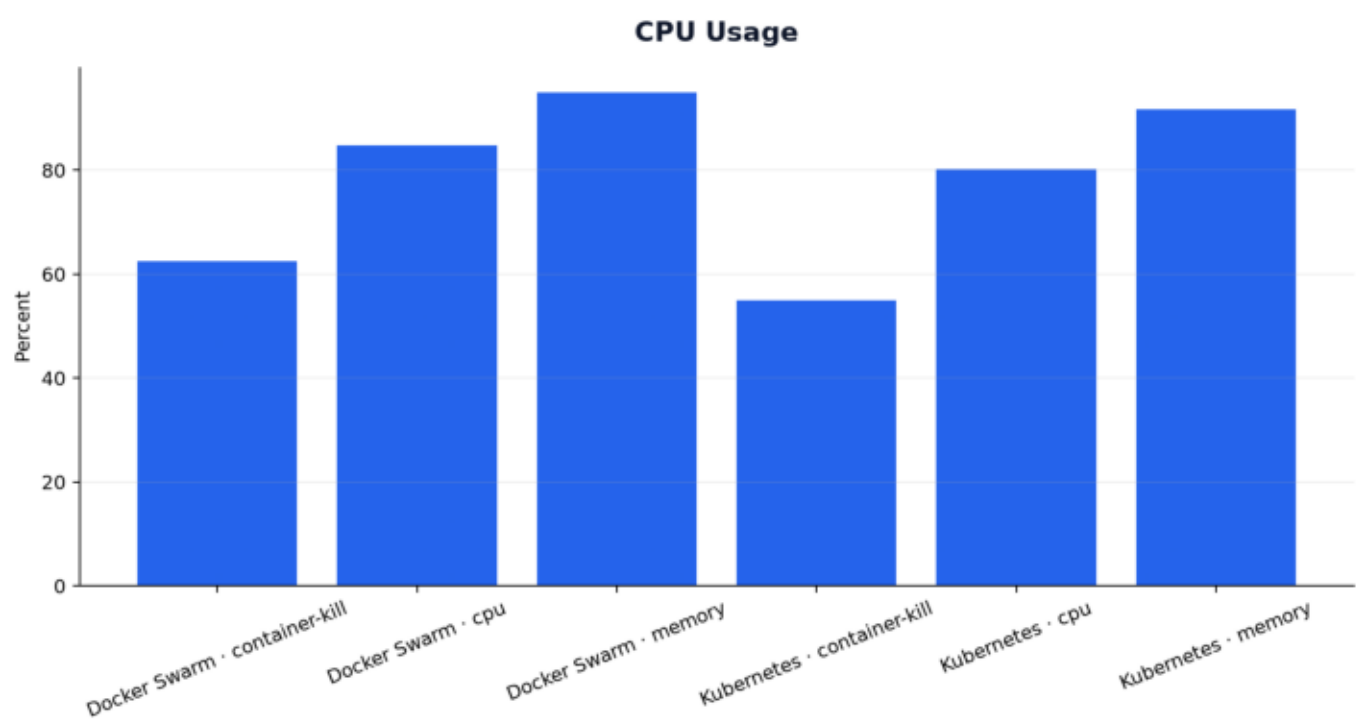
Availability

Current-session measured data



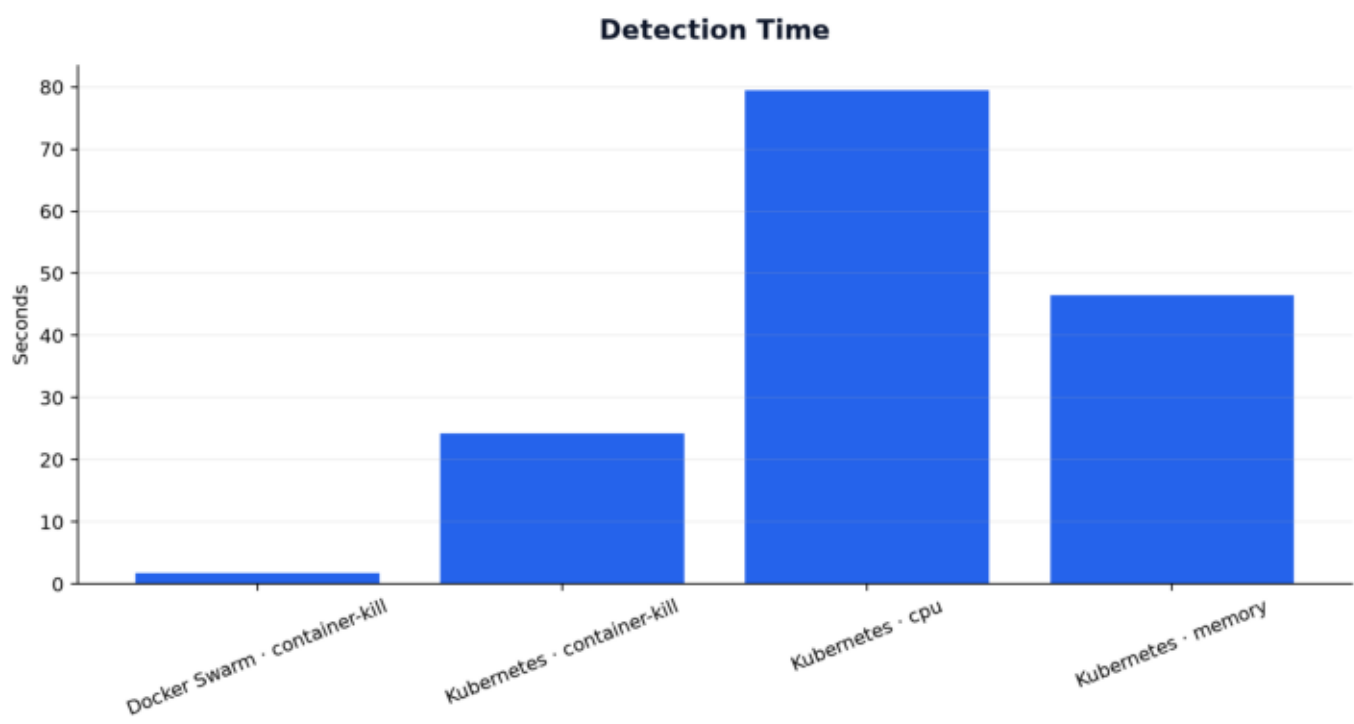
Cpu Usage

Current-session measured data



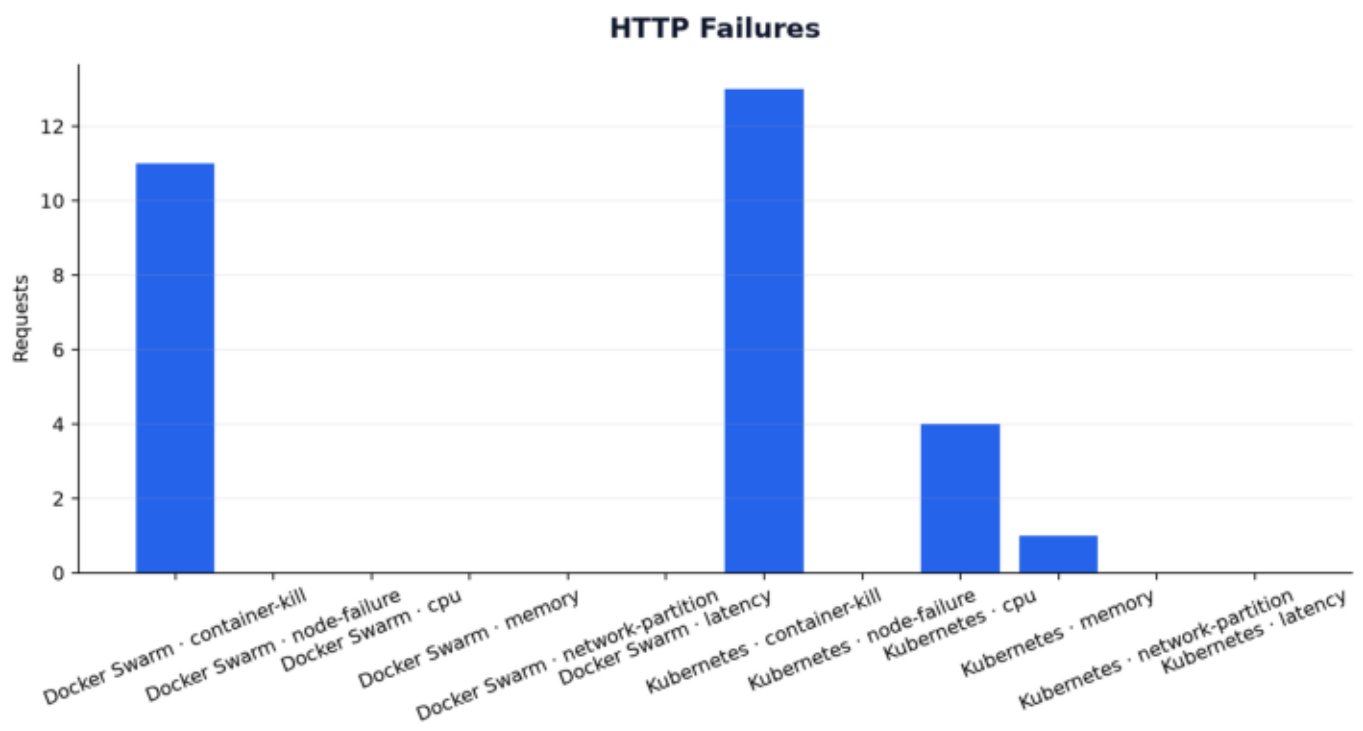
Detection Time

Current-session measured data



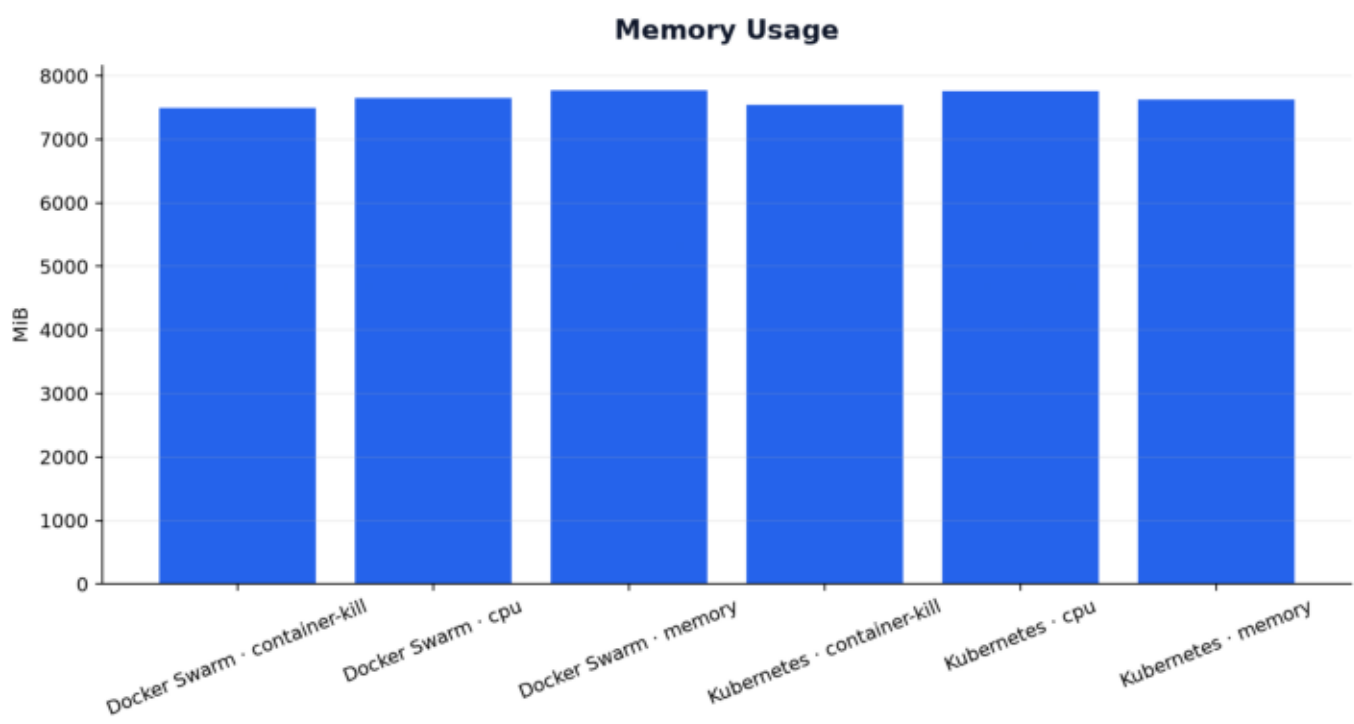
Http Failures

Current-session measured data



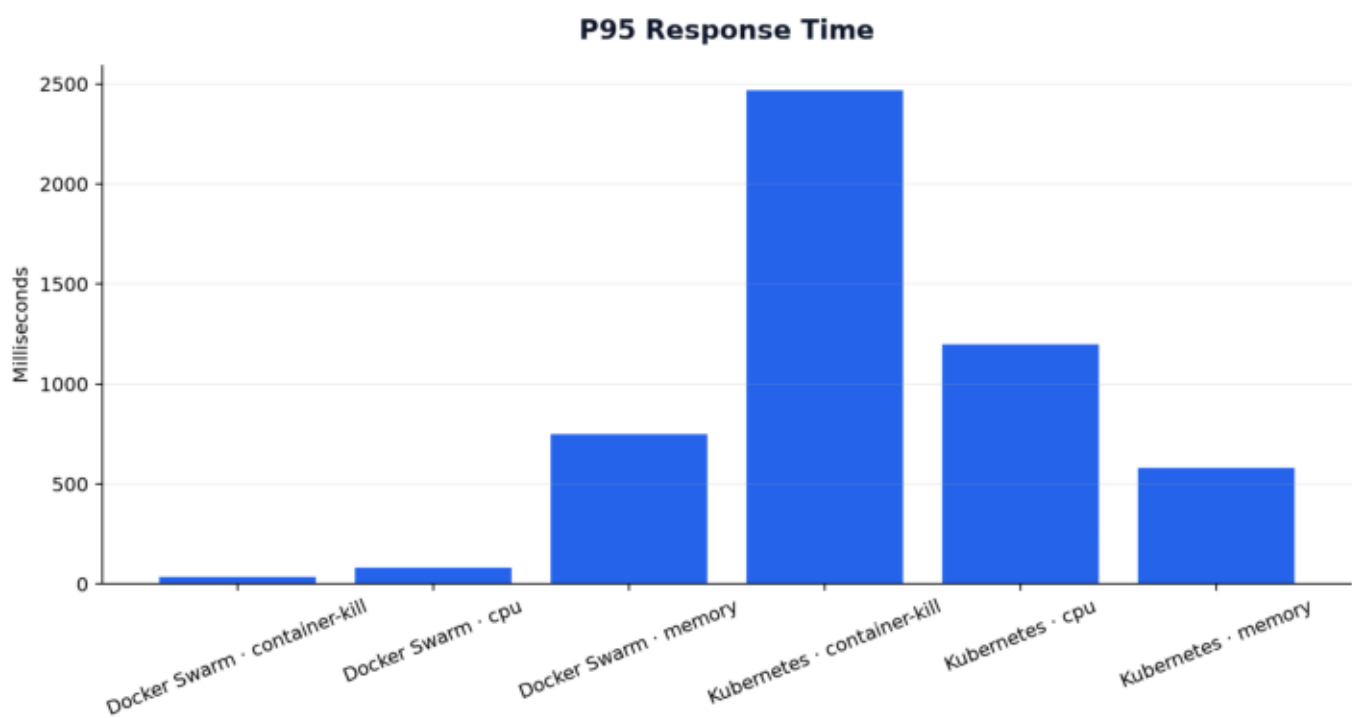
Memory Usage

Current-session measured data



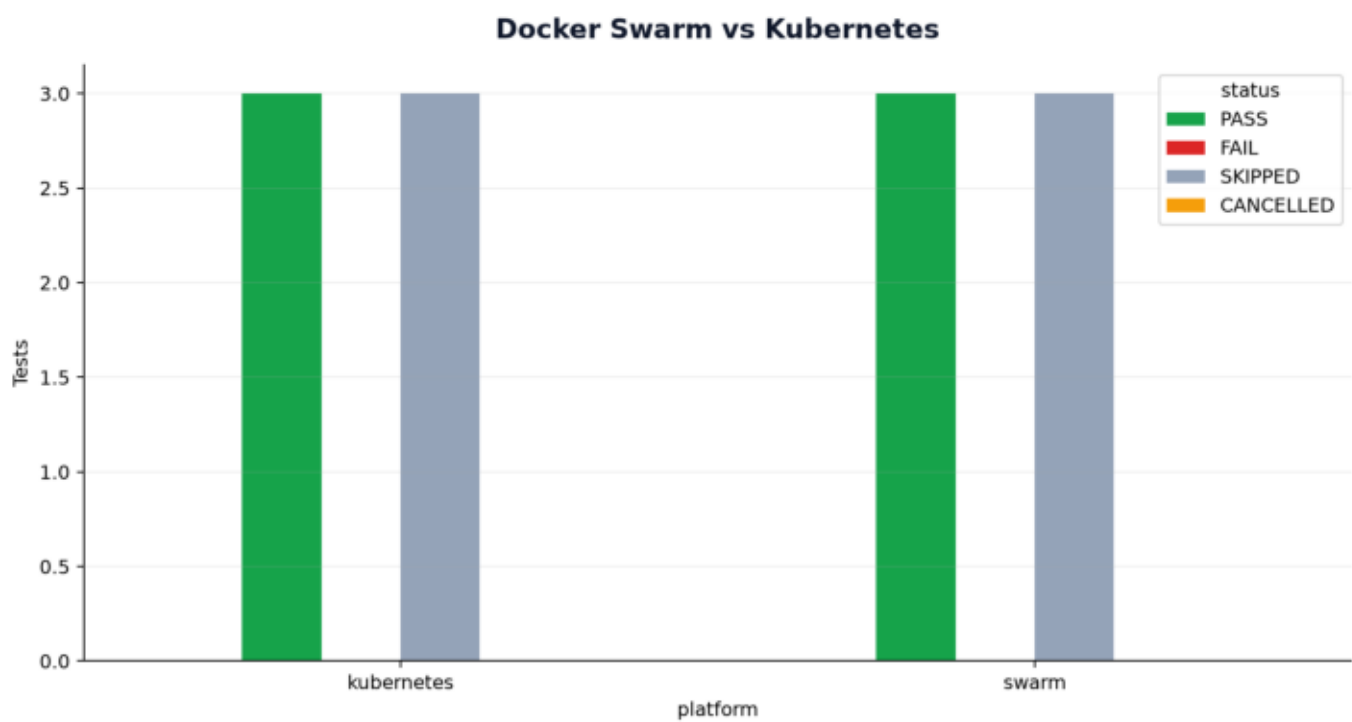
P95 Response Time

Current-session measured data



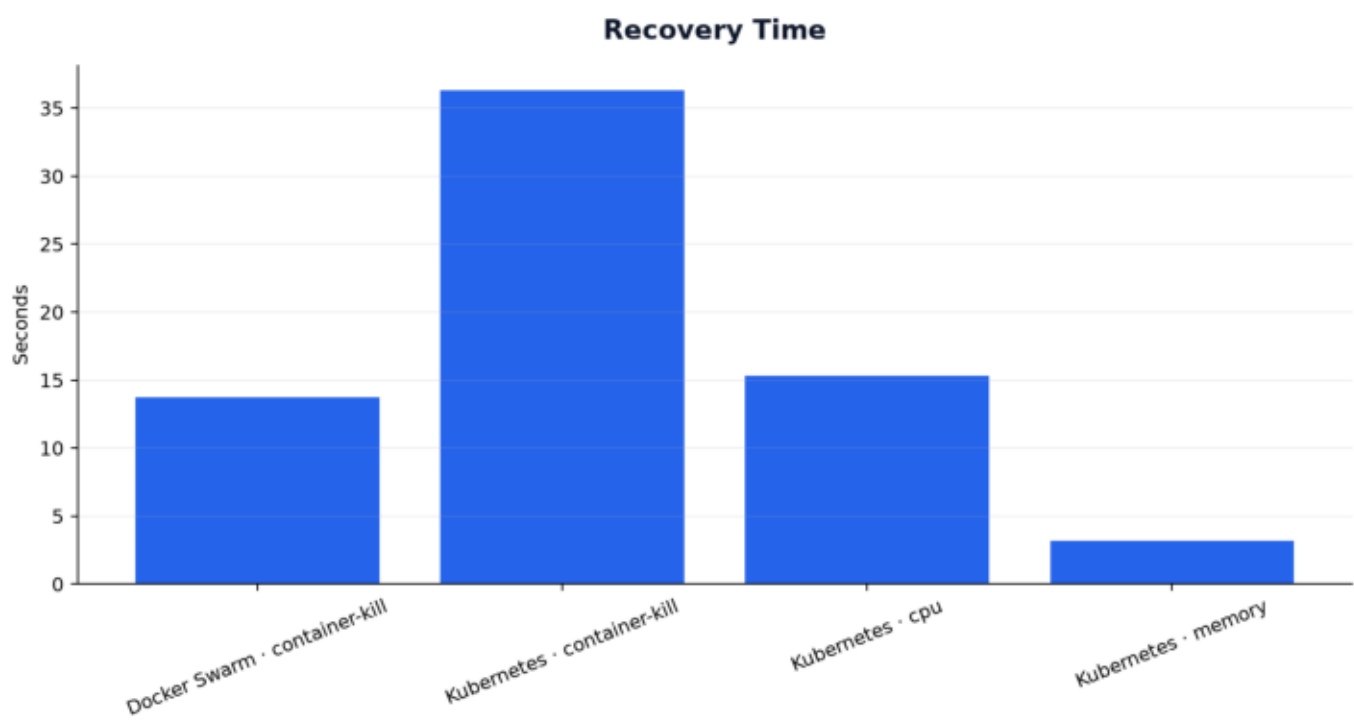
Platform Comparison

Current-session measured data



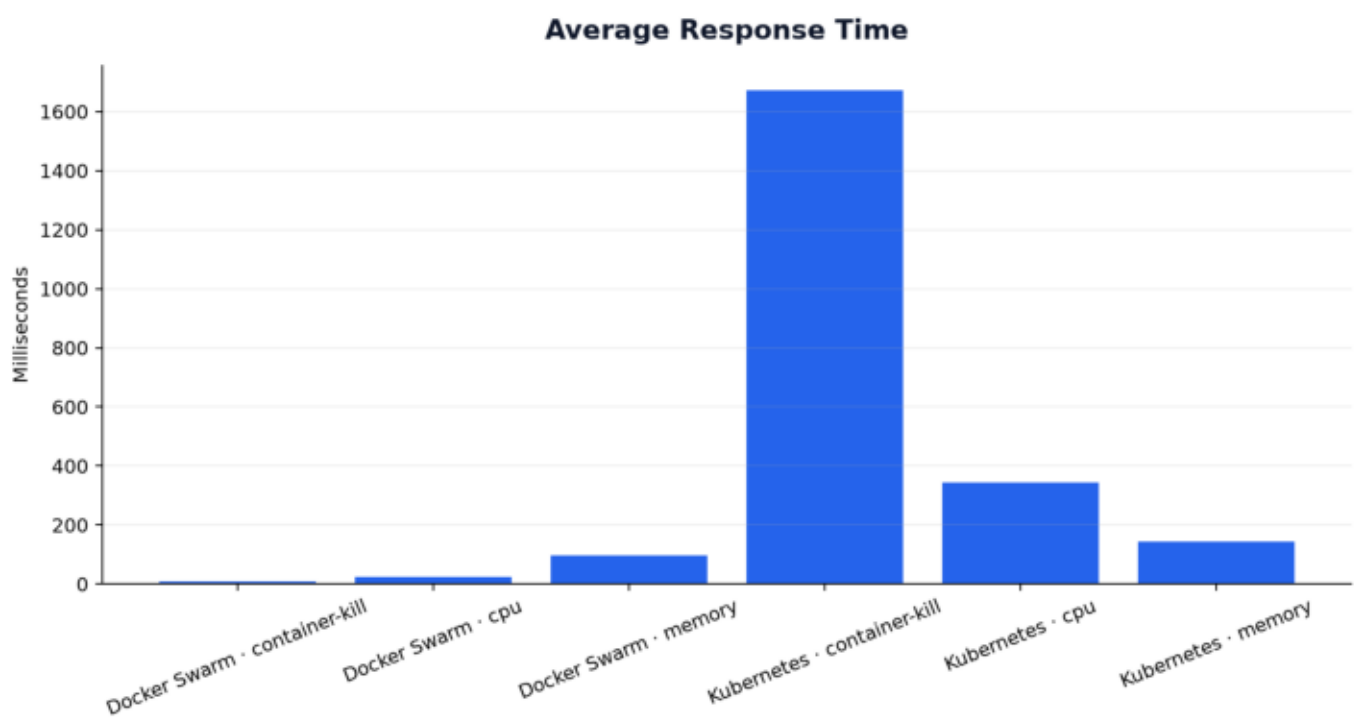
Recovery Time

Current-session measured data



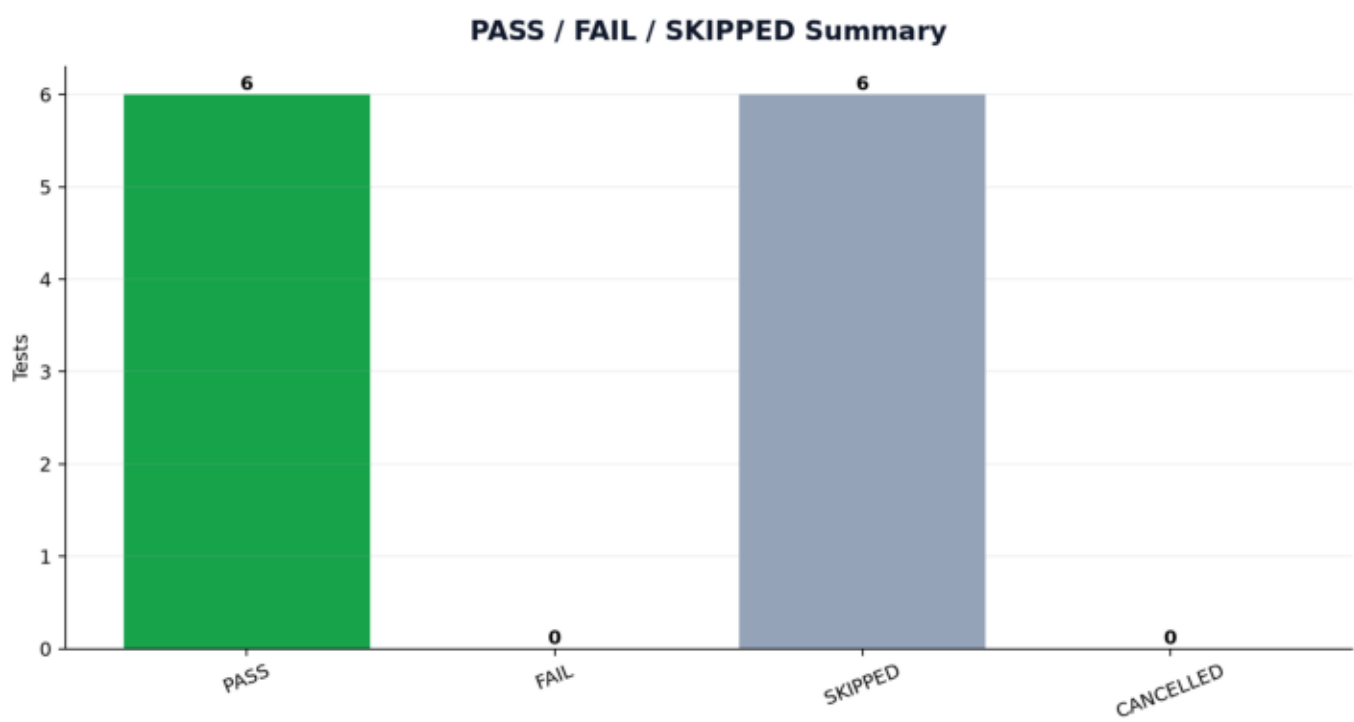
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
—	—	—	No failures recorded.

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling. Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
swarm	network-partition	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
swarm	latency	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	node-failure	Minikube inter-node CN/DNS does not provide Database probe routes. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	network-partition	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	latency	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

Swarm vs Kubernetes Comparison

Platform	PASS	SKIPPED
kubernetes	3	3
swarm	3	3

Final Conclusion

20260813_213606_4ddcf2

During this campaign, 6 of 12 scenarios passed, 0 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.